

*Teaching Notes 3B**

Bond Valuation

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Overview

- What Are Bonds?
 - Kinds and Types Of Bonds
- Default Risk and Bond Ratings
- Pricing of Bonds
 - Bond Yields
- The Term Structure of Interest Rates
- Duration and Maturity
- Appendix
 - How Active Is the Bond Market In The US?
 - Additional points on Zero Coupon Bonds
 - Implied Forward Rates
 - Volatility, Duration and Convexity

Objectives

- Understand bond pricing and probability of defaults of different fixed income securities.
- Talk about the **Yield Curve** which is also known as “**The Term Structure of Interest Rates**” from pure discount bonds and from coupon-bearing bonds.

What are Bonds?

What Are Bonds?

- A **bond** is a security that is issued in connection with a borrowing arrangement.
 - The arrangement obligates the issuer to make specified payments to the bondholder on specified dates.
- A typical coupon bond obligates the issuer to make semi-annual payments of interest to the bondholder for the life of the bond.
 - These are called **coupon payments**.
 - The coupon rate of the bond serves to determine the interest payment.
- When the bond matures, the issuer repays the debt by paying the bondholder the bond's par value (or face value).
- Bond prices adjust so that expected total returns are **similar** across bonds with similar risk, maturity, and liquidity.

Types Of Bonds: Coupon Bonds

- Constant fixed coupon (interest) payment, c .
 - Quoted as % of face value.
 - Lump-sum return of face value at maturity.
 - The price of the bond is the PV of coupon payments.
 - Issue is often at par = Face Value, F .
 - The common frequency is annual or semi-annual.
- The return on a coupon bond is given by: $Return = \frac{P_1 - P_0 + c \times F}{P_0}$
 - Investors care about the total return, which is the sum of the two components.

$$Capital\ Gain = \frac{P_1 - P_0}{P_0}$$

$$Current\ Yield = \frac{c \times F}{P_0}$$

Bond Prices: Floating Rate and Fixed Rate

- **Floating Rate Bonds:**

- Multi-period bonds.
- Pay an interest rate that is reset each period according to an index (*e.g.* LIBOR, Treasury).
 - **Example:** 3-year floating rate bond at 1 year, LIBOR + 1%
- Instrument with growing popularity.

- **Fixed Payment Bonds:**

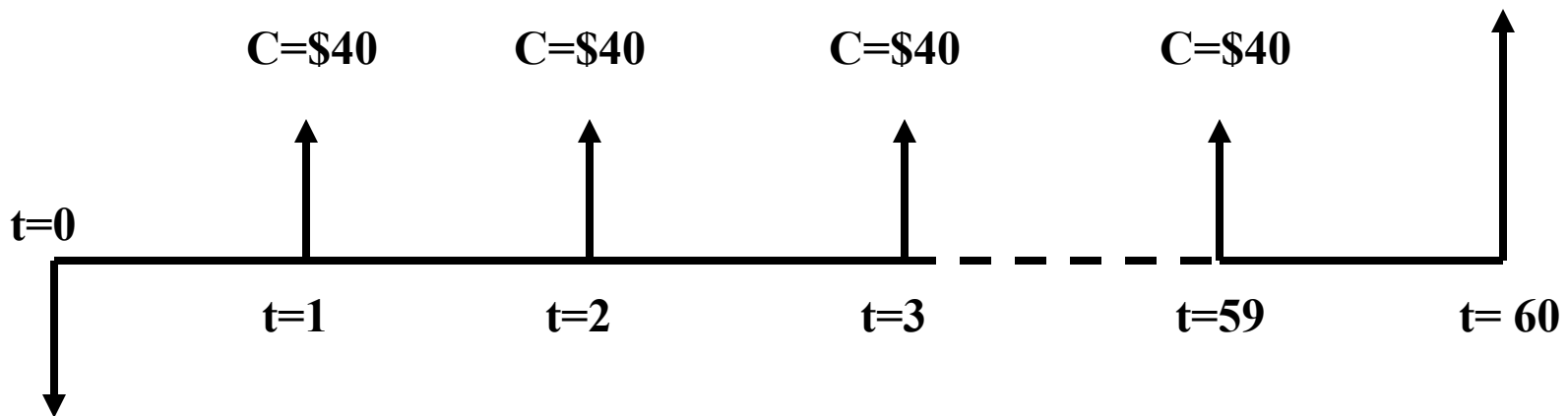
- Constant payments with gradual amortization.
 - **Example:** Mortgage Loans, Auto Loans.

Two Particular Types Of Bonds

Semi-Annual Payments

- Bond with par value of \$1,000, annual coupon rate of 8% (semi-annual payments), and maturity 30 years.
- The bondholder is entitled to semi-annual payments of \$40 ($=0.08/2 \times 1,000$) for the life of the bond, plus \$1,000 par value at maturity.

$F + cF = \$1,040$



- **Bottom Line:** Bond Price = $B(C, T) = B(4\%, 60)$

Two Particular Types Of Bonds

Semi-Annual Payments: Example

US Treasury 3.125% 05/15/2048

Retrieved: 11:38:07 AM ET 10/20/2018

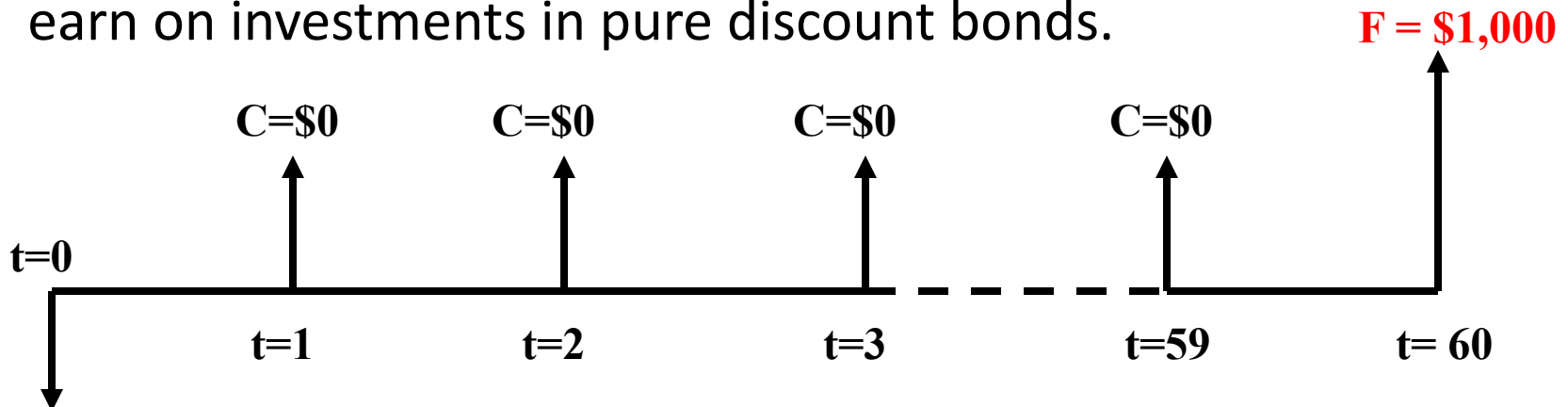
Security Details		Quote Details	Bid	Ask
CUSIP	912810SC3	Price	95.234	95.320
Security Type	U.S. Treasuries	Current Yield	3.281%	3.278%
Maturity Date	05/15/2048	Yield To Maturity	3.381%	3.376%
Coupon Rate	3.125%	Yield To Call	--	--
Coupon Type	Fixed	Yield To Worst	--	--
Coupon Frequency	Semi-annually	Available Quantity	5000	5000
Accrual Day Count	ACT/ACT	Trade Min/Increment	25 / 1	25 / 1
Dated Date	05/15/2018	Security Min/Increment	1 / 1	1 / 1
First Settlement Date	--	Price Details	Price based on 25 bonds and a settlement date of 10/23/2018	
First Coupon Date	11/15/2018			

- 30-year US Treasury bond, 3.125% semi-annual fixed coupon

Two Particular Types Of Bonds

Zero-Coupon Bonds

- Zero-Coupon Bonds or **Pure Discount Bonds** are bonds that pay a single cash-flow, the face value at maturity.
- The prices of these bonds (relative to their face value) are direct indicators of the discounting or the “pure discounting”.
 - The prices of these bonds are determined in a market.
 - From these prices, we “back-out” the returns that investors earn on investments in pure discount bonds.



Two Particular Types Of Bonds

Zero-Coupon Bonds: Example

US Treasury STRIP 0% 05/15/2048

Retrieved: 11:47:21 AM ET 10/20/2018

Security Details		Quote Details	Bid	Ask
CUSIP	912803FD0	Price	36.751	37.225
Security Type	U.S. Treasury Zeros	Current Yield	0.000%	0.000%
Maturity Date	05/15/2048	Yield To Maturity	3.415%	3.371%
Coupon Rate	0.000%	Yield To Call	--	--
Coupon Type	Zero	Yield To Worst	--	--
Rate Schedule	08/02/2018 - 0.000	Available Quantity	5000	5000
Coupon Frequency	--	Trade Min/Increment	25 / 1	25 / 1
Accrual Day Count	ACT/ACT	Security Min/Increment	1 / 1	1 / 1
Dated Date	05/15/2018	Price Details	Price based on 25 bonds and a settlement date of 10/23/2018	

- 30-year zero-coupon US Treasury bond, 0% coupon
- Notice that the price is much lower than for the bond with semi-annual coupons!

What Kind Of Bonds Are Being Traded?

- **Treasury Bonds** (maturity between 10-30 years), **Treasury Notes** (maturity < 10 years), **Treasury Bills** (maturity < 1 year): Issued by the government.
- **Corporate Bonds:** Issued by corporations.
- **Callable Bonds:** Call provision allows the issuer to repurchase the bond at a specified price before the maturity date.
- **Convertible Bonds:** Give bondholders an option (the right but not the obligation) to exchange each bond for a specified number of shares of stock of the firm.

What Kind Of Bonds Are Being Traded?

- **International Bonds:** Dollar denominated foreign bonds sold in the United States are called **Yankee Bonds**.
- **International Bonds:** The Eurodollar market refers to dollar-denominated bonds sold outside the United States (not just in Europe).
- **Indexed Bonds** make payments that are tied to a general price index or the price of a particular commodity (*e.g.*: Inflation-indexed bonds, “Morgan Stanley” TIPS).
- **Remark: Seniority** is the priority with which bonds have access to the firm’s assets in the event of bankruptcy.

Default Risk And Bond Ratings

Default Risk and Bond Rating

- Although bonds generally **promise** a fixed flow of income, income is not risk-free unless the investor can be sure the issuer will not default on its obligation.
- While US government bonds can be treated as free of default risk, r_f (risk-free rate), this is not true of corporate bonds.
 - If the company goes bankrupt, the bondholders will not receive all the payments they have been promised. The actual payments are uncertain, for they depend to some degree on the ultimate **financial status** of the firm.
- Bond default risk is measured by Moody's Investor Services (www.moodys.com) and Standard & Poor's Corporation (www.standardandpoors.com) among others.

Default Risk and Bond Rating

<u>Moody' s</u>	<u>Standard & Poor's</u>	<u>Safety</u>
Aaa	AAA	The strongest rating; ability to repay interest and principal is very strong.
Aa	AA	Very strong likelihood that interest and principal will be repaid
A	A	Strong ability to repay, but some vulnerability to changes in circumstances
Baa	BBB	Adequate capacity to repay; more vulnerability to changes in economic circumstances
Ba	BB	Considerable uncertainty about ability to repay.
B	B	Likelihood of interest and principal payments over sustained periods is questionable.
Caa	CCC	Bonds in the Caa/CCC and Ca/CC classes may already be in default or in danger of imminent default
Ca	CC	
C	C	C-rated bonds offer little prospect for interest or principal on the debt ever to be repaid.

Default Risk and Bond Rating

	Moody's	S&P's & Fitch
Investment Grade		
	Aaa	AAA
	Aa	AA
	A	A
	Baa	BBB
Junk Bonds/High Yield Bonds		
	Ba	BB
	B	B
	Caa	CCC
	Ca	CC
	C	C

- **Junk Bonds** are lower rated bonds for which default risk is high and the inherent risk of the company is high!
 - Junk or High Yield Bonds are also used for Speculative Trading.
 - The market for this type of issue is **very cyclical**

Default Risk and Bond Rating

- **Determinants** of Bond Safety:
 - Coverage ratios: ratios of earnings to fixed costs.
 - Leverage ratio: debt-to-equity ratio.
 - Liquidity ratios: current assets/current liabilities and current assets excluding inventories/current liabilities.
 - Profitability ratios: return on assets.
 - Cash flow-to-debt ratio.

Three-years of Median Ratio Data By Bond Rating (2010-2013)

Ratio	AAA	AA	A	BBB	BB	B	CCC
EBIT Interest Cover *	21.4	10.1	6.1	3.7	2.1	0.8	0.1
Return on Capital %	34.9	21.7	19.4	13.6	11.6	6.6	1
Gross Profit Margin %	27	22.1	18.6	15.4	15.9	11.9	11.9
Total Debt/Capital %	22.9	37.7	42.5	48.2	62.6	74.8	87.7
* Earnings before interest and tax divided by interest							

Default Risk and Bond Rating

- Few people know the “weights” applied to banks’ financial health ratios when determining their credit ratings.
 - Can try to estimate them.
 - Credit ratings for banks tend to give a high weight to coverage ratios.

Ratio Data for US Banks (Q2 2018)

Ratio	Bank of America	Citigroup	Wells Fargo	PNC Bank	JPMorgan Chase
Quick Ratio %	35%	52%	32%	33%	19%
Interest Coverage	3.1	2.4	3.7	4.2	3.6
Debt Coverage %	19%	17%	13%	13%	19%
Leverage Ratio	7.7	8.2	8.1	7.3	9.1
Debt to Equity	1.0	1.3	1.6	1.4	1.3
Moody's Rating	A	Baa	A	A	A

Default Risk and Bond Rating

**Default Rates of Corporate Bonds
by S&P's Rating at Time of Issue (1981-2017 sample)**

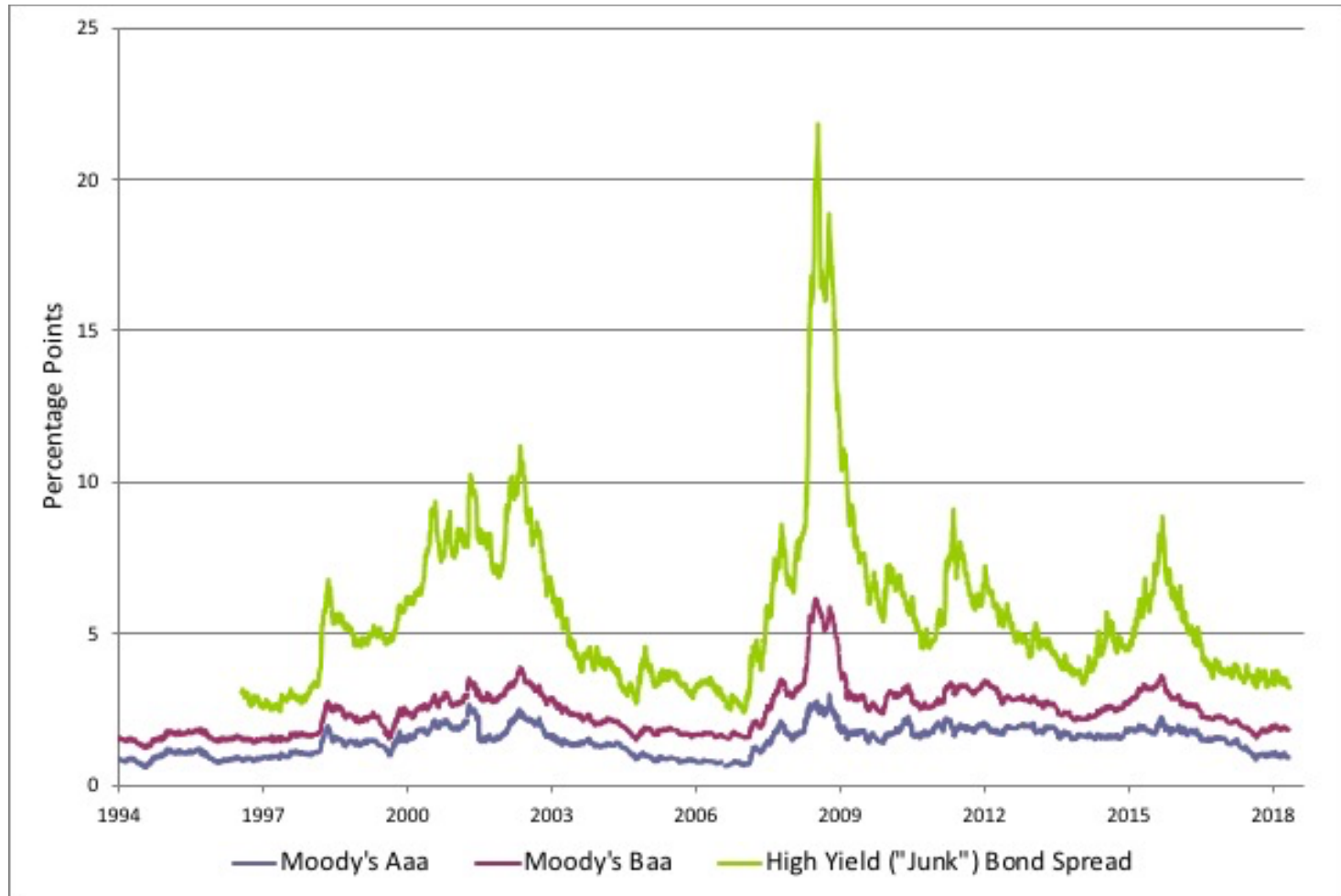
	Percentage Defaulting Within:		
Rating at Time of Issue	1 Year After Issue	5 Years After Issue	10 Years After Issue
AAA	0.0	0.4	0.7
AA	0.0	0.4	0.9
A	0.1	0.5	1.5
BBB	0.2	1.4	3.2
BB	0.6	6.5	11.5
B	3.6	17.9	24.8
CCC-C	26.8	46.2	50.7
All Rated	1.5	6.2	9.1

Example: Tesla (March 27, 2018)

- S&P cut the B3 long-term corporate credit rating of Tesla to B2. Tesla is assigned a **Negative Outlook**.
- **Yield Spreads** on Tesla's 5.3% notes due in 2025 widened to 4.46% points **more** than 7-year Treasuries, up from 3.8% points the previous day.
- Tesla's rating reflects a shortfall in production of the company's Model 3 electric vehicle. The company also faces liquidity pressures from large negative FCF and the pending maturities of convertible bonds (\$230 M in November 2018, \$920 M in March 2019).
- **Note:** Yield difference (or spread) is the credit spread between Corporate and Government Bonds of similar maturity.
 - It is normally measured in basis points.

Bond Ratings And Yield Spreads

Yield Spreads Between Corporate And 10-year Treasury Bonds



Bond Ratings And Yield Spreads

- When the economy is very good, a company that is strong financially will need to offer only a very small yield spread over treasuries.
 - In 2012, IBM offered just a few basis points more than 10-year Treasuries to attract buyers to their bonds.
- If an issuer has credit problems, the yield spread of its bonds over treasuries will widen.
 - Tesla's yield spreads **widened** dramatically when they announced production shortfalls in the context of tight liquidity.
 - They could no longer issue debt at 3.8% over the 7-year T-rate.
- **Bottom Line:** Investors demand higher yields to **compensate** for higher risk.

Bond Pricing

Bond Pricing

- Let's start by focusing on Government Bonds.
 - In other word, bonds with **no** default risk or risk-free rate bonds.
- To value a security we discount its cash-flows by the appropriate discount rate.

$$\text{Bond value} = \text{PV}(\text{Coupons}) + \text{PV}(\text{Face Value})$$

- Define maturity date T and discount rate r , the value of the bond V is given by:

$$B(C, T) = \sum_{t=1}^T \frac{C_t}{(1+r)^t} + \frac{F}{(1+r)^T}$$

Bond Pricing

Example

- 8% coupon, 30-year maturity bond, par value of \$1,000, semi-annual coupon payments.
- Suppose that the discount rate is 8% annually, or 4% per six-month period. The value of the bond:

$$B(40,60) = \sum_{t=1}^{60} \frac{\$40}{(1 + 0.04)^t} + \frac{\$1,000}{(1 + 0.04)^{60}} = \$1,000$$

- **Note:** The coupon rate equals the discount rate and the bond price **equals** par value.
- **Question:** What if the discount/interest rate is higher/lower?

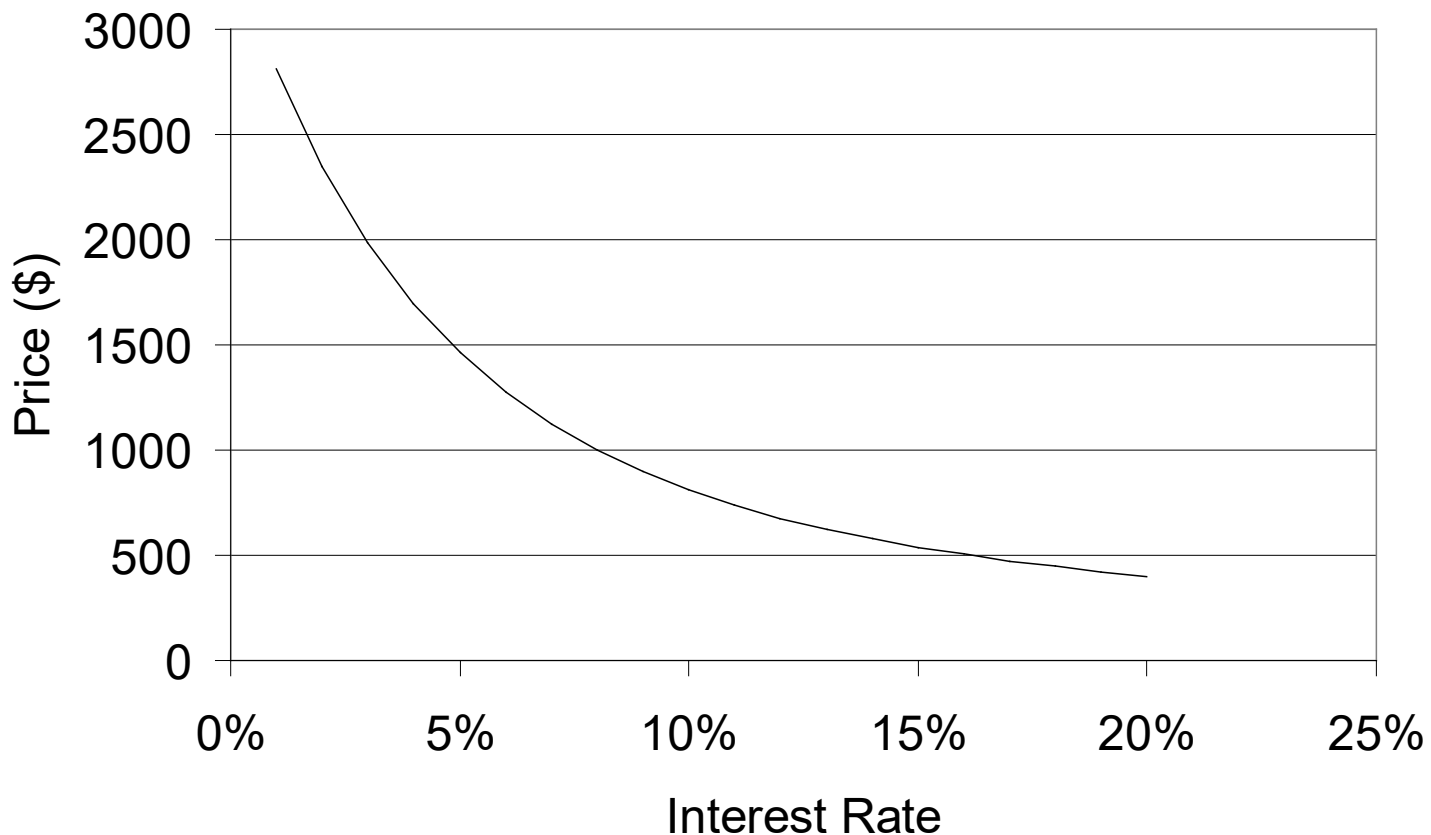
Bond Pricing

Market Interest/Discount Rate	Bond Price
4%	\$ 1,695.22
6%	\$ 1,276.76
8%	\$ 1,000.00
10%	\$ 810.71
12%	\$ 676.77

- At a higher interest rate, the present value of the payments to be received by the bondholder is smaller. Therefore the bond price will **fall** as market interest rates **rise**.
 - This illustrates a crucial general rule in bond valuation.

General Rule In Bond Valuation

- Price of the 30-year, 8% coupon bond for a range of interest rates:



General Rule In Bond Valuation

- The negative slope illustrates the **negative** relationship between prices and interest rates, which is a general feature of fixed-income securities.
 - If you buy a bond at par for 8% and market rates subsequently rise, then you suffer a loss.
 - You have tied up your money earning 8% when alternative investments earn higher returns.
- The shape of the curve also implies that an increase in interest rates results in a price decline that is smaller than the price gain resulting from a decrease of equal magnitude in the interest rate.
 - This property of bond prices is called **convexity** because of the convex shape of the bond price curve.

Bond Yields: Current/Interest Yield

- **Current Yield** or **Interest Yield** is equal to the cash income provided by the bond as a percentage of its price.

$$\text{Interest Yield} = \frac{\text{Interest}}{\text{Price}} = \frac{\text{Coupon}}{\text{Price}}$$

- **Example:** An 8% coupon, face value \$1,000, 30-year bond is selling at \$1,276.76.
- The current yield in annual terms is:

$$\$80/\$1,276.76 = \mathbf{6.27\%}$$

- The current yield **ignores** any prospective capital gains or losses.

Bond Yields: Yield To Maturity

- **Yield To Maturity** (YTM) or **Redemption Yield** is the interest rate that makes the present value of a bond's payments equal to its price.
 - This is basically the internal rate of return of the bond!
- **Question:** In our previous example, what is the YTM?

$$\$1,276.76 = \sum_{t=1}^{60} \frac{\$40}{(1 + YTM)^t} + \frac{\$1,000}{(1 + YTM)^{60}}$$

- **YTM ~ 3%**
- The YTM is the average rate of return earned by an investor purchasing the bond at \$1,276.76 **and** holding the bond to maturity.

What About Default Risk?

- The yield on a bond represents the return if all payments are made.
 - This may not be true due to **default risk**.
- The price of a bond with default risk is equal to the **expected** (not promised) payments, discounted at an appropriate risk-adjusted rate.
 - An issue that will confront us later

The Term Structure Of Interest Rates

What Is The Yield Curve?

- A line that plots the interest rates, at a set point in time, of bonds having **equal** credit quality but **differing** maturity dates.
 - The time-0 spot yield curve, $\mathbf{Y}_0 = \{r_0(1), r_0(2), \dots, r_0(T), \dots\}$, is the list of rates currently available on risk-free investments with different horizons.
- The most frequently reported yield curve compares the 3-month, 2-year, 5-year and 30-year U.S. Treasury debt.
 - This yield curve is used as a **benchmark** for other debt in the market, such as mortgage rates or bank lending rates.
 - The curve is also used to **predict** changes in economic output and growth.

Why Worry About The Yield Curve?

- The yield curve plots the interest rate that you can borrow/lend as a function of the borrowing/lending horizon.
- If the yield curve is **not** flat:
 - You cannot discount all cash flows at the same interest rate.
 - Therefore, the yield curve is needed for valuation purposes.
- If the yield curve **changes** over time, then:
 - Your cost of borrowing changes over time.
 - Forward interest rate contracts can be used to lock in interest rates and therefore future cash flows.
 - Shifts in the yield curve changes the value of a cash flow stream.
- **Question:** What does the Yield Curve look like **nowadays**?
 - Check <http://www.smartmoney.com/onebond> or <https://www.marketwatch.com/investing/bond/tmubmusd10y?countrycode=bx>

What Drives The Yield Curve?

- A major factor affecting interest rates is investors **expectations for inflation** and **economic growth**.
- In August 2006:
 - Fed Funds rate (overnight) = 5.25%
 - 10-year Treasury Bond = 5%
 - **Question:** What does this mean?
- The downward sloping yield curve says that inflation will not erode the value of that 5% yield.
 - Otherwise investors would require a greater expected return to lend for 10 years.
- Expectations of future economic growth have an important influence on interest rates.

The Term Structure Of Interest Rates

- The term structure of interest rates is the set of yields to maturity, at a given time, on zero-coupon bonds of different maturities.
- To construct a spot yield curve with coupon bearing bonds, treat each bond as a portfolio of its component parts.
 - In other words, decompose the total bond value today into the value of each coupon or principal payment today.

When Is The Yield Curve Normal?

- **Normal Yield Curve:**

Through most of the post-Great Depression era to present the yield curve has been called **normal** when yields rise as maturity lengthens, that is, when the slope of the yield curve is positive.

This **positive slope** reflects investor expectations for the economy to grow in the future and, importantly, for this growth to be associated with a greater risk that inflation rises in the future than falls.

This expectation for higher inflation in the future than the present generates both an expectation that the central bank will tighten monetary policy by raising short term interest rates in the future to slow economic growth and dampen inflationary pressure and the need for a risk premium associated with the uncertainty about the future rate of inflation and the risk this poses to the future value of cash flows. Investors price these risks into the yield curve by demanding higher yields for maturities further into the future.

When Is The Curve Flat Or Inverted?

- **Flat Yield Curve:**

A **flat** curve is apparent when all maturities have the same yield. A **flat** curve sends signals of uncertainty into the economy. This mixed signal can revert back to a normal curve or could later result in an inverted curve.

- **Inverted Yield Curve**

An **inverted** curve occurs when long-term yields fall below short-term yields. Under this abnormal and contradictory situation, long-term investors will settle for lower yields now if they think the economy will slow or even decline in the future. An inverted curve may indicate a worsening economic situation in the future. In addition to potentially signaling an economic decline, inverted yield curves also imply that the market believes inflation will remain low. This is because, even if there is a recession, a low bond yield will still be offset by low inflation.

Duration And Maturity

Duration and Maturity

- We already saw that the price of a bond depends on interest rates. Portfolio managers and firms need to know how the value of their bond portfolio responds to **changes** in interest rates and then manage this sensitivity.
- A bond's maturity does **not** take into account the arrival of coupons prior to maturity.
- Duration (**D**) is a measure of the average maturity of the bond's promised cash flows.
 - Is a measure of time horizon that takes into account **all** cash flows.
 - It also provides a measure of the **sensitivity** of a bond to interest rate changes.
 - It is also known as the **effective maturity** of a bond.

Duration and Maturity

- Frederic Macaulay (1938) suggested that duration be computed as the **weighted** average of the times to each coupon or principal payment made by the bond.
 - The weight applied to each payment is the proportion of the total value of the bond accounted for by that payment. The weight **w_t** associated with the cash flow made at time t (denoted CF_t) would be:

$$w_t = \frac{PV(CF_t)}{\text{Bond Price}} = \frac{CF_t / (1 + y)^t}{\text{Bond Price}}$$

- The duration is:

$$D = \sum_{t=1}^T t \times w_t = \frac{1 \times PV(CF_1)}{\text{Bond Price}} + \frac{2 \times PV(CF_2)}{\text{Bond Price}} + \dots + \frac{T \times PV(CF_T)}{\text{Bond Price}}$$

Duration and Maturity

- The formula weights the periods in which the cash flows arrive by the present value arriving in that period.
 - If most of the value arrives in year 2, the duration of the bond will be about 2.
 - If most of the values arrives at the end of the bond's life, then its duration will be nearer the bond's maturity.
- **Example:** A bank finances a commercial real estate development with a 5-year loan, with annual payments of \$2M, a final payment of \$15M, and a yield of 12%. Compute the **duration** of the loan.

Example

- The **Loan Price** is given by:

$$P = \frac{2}{(1.12)} + \frac{2}{(1.12)^2} + \frac{2}{(1.12)^3} + \frac{2}{(1.12)^4} + \frac{17}{(1.12)^5}$$
$$= 1.786 + 1.594 + 1.424 + 1.272 + 9.646 = 15.72$$

- The **Duration** is therefore:

$$D = \frac{1 \times 1.786}{15.72} + \frac{2 \times 1.594}{15.72} + \frac{3 \times 1.424}{15.72} + \frac{4 \times 1.272}{15.72} + \frac{5 \times 9.646}{15.72}$$
$$= (1 \times .114) + (2 \times .101) + (3 \times .091) + (4 \times .081) + (5 \times .614)$$
$$= 3.983$$

- **Question:** Is there any relationship between **volatility** and **duration**?

Four Facts About Duration

- There are **four general facts** about duration
 1. **Maturity:** A longer maturity increases the duration of a bond (for a given coupon rate and yield).
 2. The duration of a coupon bond is always **less** than its maturity.
 3. **Coupon Rate:** A higher coupon rate decreases the duration of a bond (for a given yield and maturity).
 4. **Yield:** A higher yield decreases the duration of a bond (for a given coupon rate and maturity).

Volatility and Duration

- Duration is useful mainly since the percentage change in the price of a bond due to a change in the bond's yield to maturity is proportional to the duration times the change in the yield:

$$\% \text{ Change in Bond Prices} \approx -D \times \frac{\% \text{ Change in Yield (\% Points)}}{1 + Y}$$

$$\% \text{ Change in Bond Prices} \approx -D^M \times \% \text{ Change in Yield (\% Points)}$$

where:

- y is the level of the yield curve before any changes.
- D^M is called the **modified duration**. $D^M = D/(1 + y)$
- The minus sign in front of D reflects the fact that when the yield curve rises, the bond price falls.
- Remark:** The longer a bond's duration, the **greater** is its volatility.

Takeaways

- A **bond** is a security that is issued in connection with a borrowing arrangement.
 - A typical coupon bond obligates the issuer to make semi-annual payments of interest to the bondholder for the life of the bond.
 - Pricing of bonds is a present value calculation
- The following **different** terms in the context of bonds are useful: maturity, duration, and yield to maturity
- The yield curve plots the interest rate that you can borrow/lend as a function of the borrowing/lending horizon.
 - Knowing the prices of ZCB with maturities 1, 2, 3,... means we also know what the price of a risk-free bond over different horizons should be.

Appendix:

How Active is Bond Market in US?

How Active Is The Bond Market In The US?

- The dollar volume of bonds traded daily is about **ten times** that of equity markets.
 - For example, a single \$15B issue of 10 year Treasury Bonds sold in one day in 2005.
- Total debt outstanding in the US Bond Market is **\$41 Trillion** as of end-2017 (SIFMA)
 - Includes treasuries, agencies, corporate bonds, and mortgage-backed securities.
- Other major sectors include corporate high yield bonds, Eurodollar bonds, emerging markets and private placements.

Appendix: Additional Points on Zero Coupon Bonds

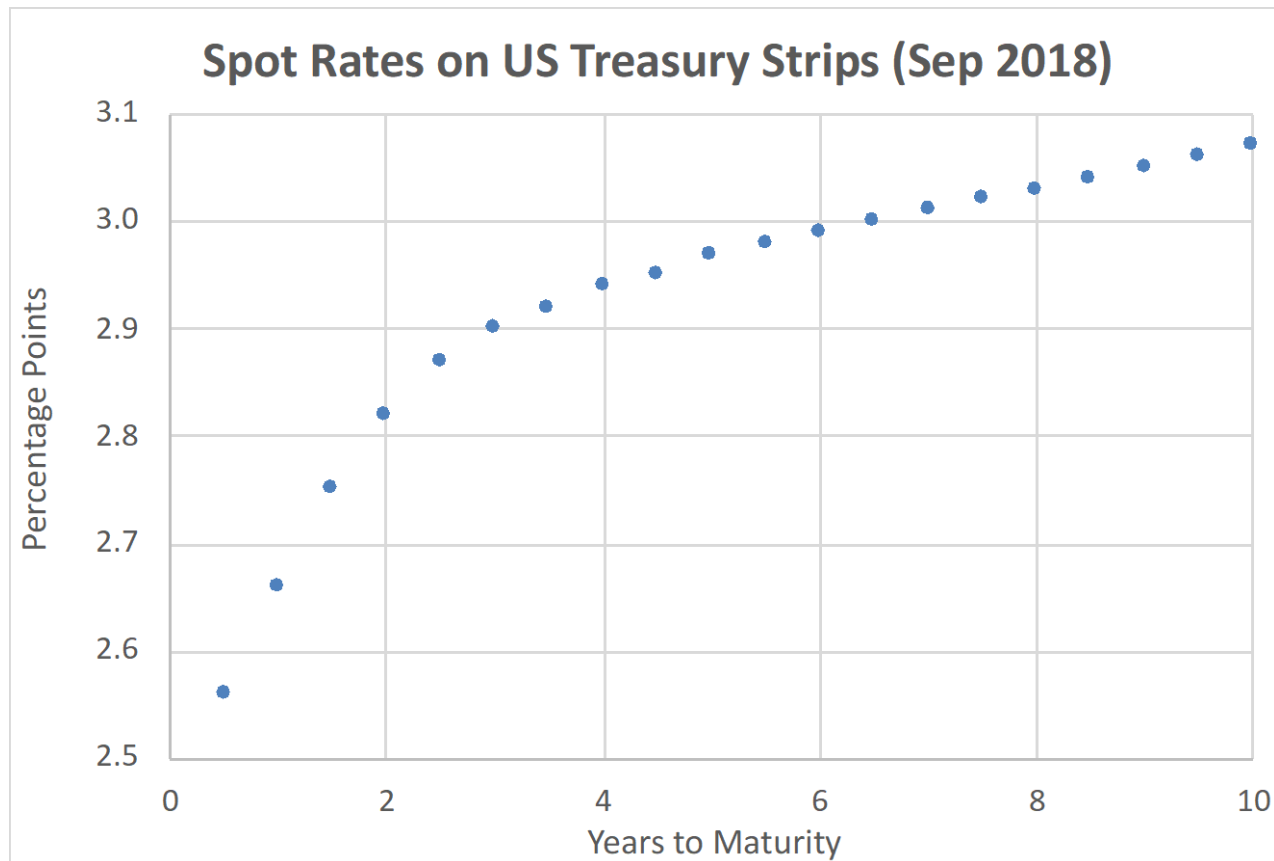
Additional Points: Zero-Coupon Bonds (ZCB)

- Recall that ZCB make just one payment at maturity.
 - Example: US T-Bills.
- The discount factor is the price today of receiving \$1 tomorrow.
 - Therefore the price of a ZCB gives immediately the **discount factor** that we should use to compute the present value of a stream of risk-free cash-flows.
- The YTM of a ZCB is the discount rate that we should use to compute the present value of a stream of risk-free cash-flows. The YTM of a ZCB is also commonly known as the **spot rate** that prevails today for a period corresponding to the maturity of the zero.

Additional Points: Zero-Coupon Bonds (ZCB)

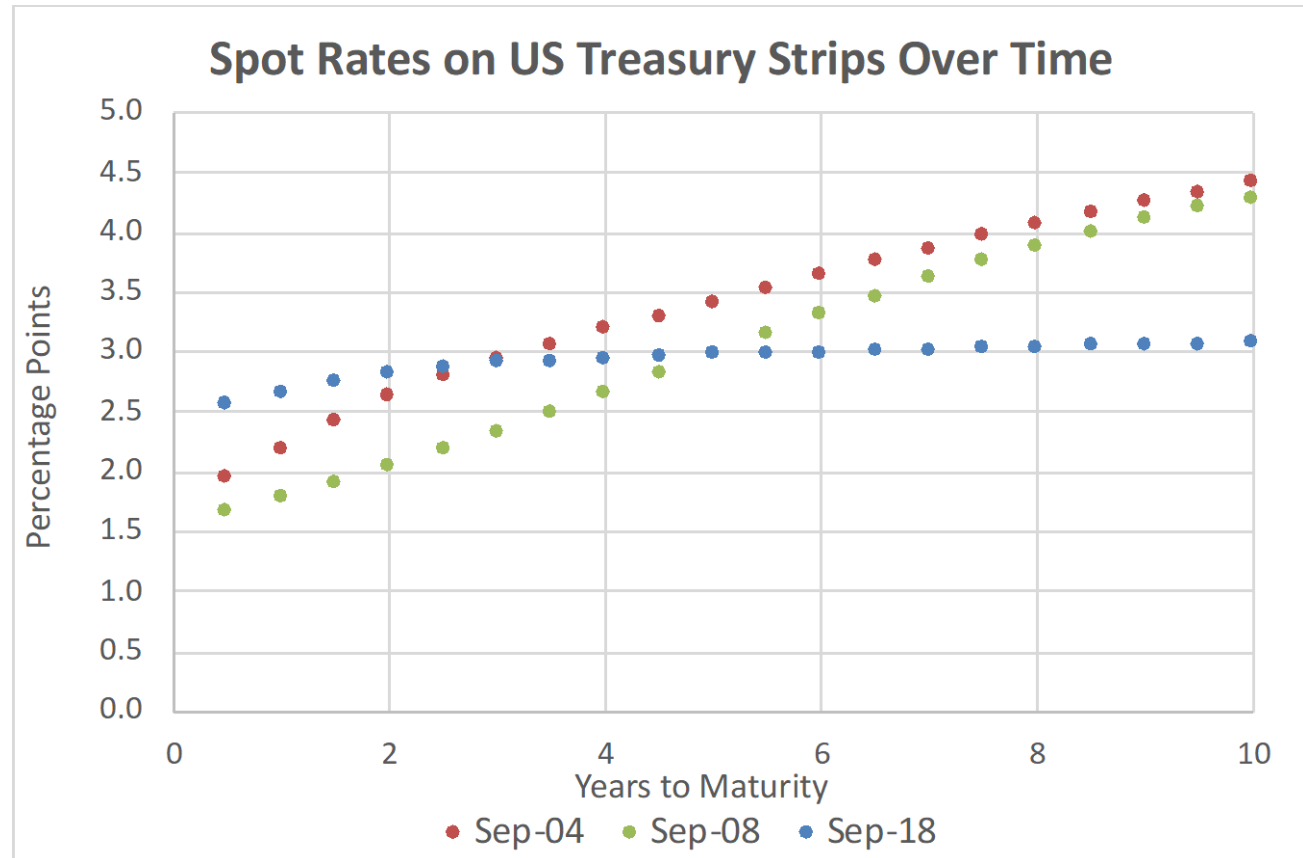
- To find the spot interest rate, we need the price of a bond that simply makes one future payment.
 - Fortunately, such bonds **do** exist!
- They are known as **stripped bonds** or just **strips**.
 - Strips originated in 1982 when several investment bankers bought US Treasury Bonds and reissued their own separate mini-bonds, each of which made only one payment.
 - The US Treasury soon began to issue its own mini-bonds.
- Let's see a plot of the term structure of spot rates from 1 to 10 years (Strips in September 2018).
 - Note the (slight) upward slope of the term structure of interest rates!

Additional Points: Zero-Coupon Bonds (ZCB)



- Investors require **2.66%** from a bond that made a payment only at the end of one year and a rate of **3.07%** from a bond that paid off only at the end of 10 years.

Additional Points: Zero-Coupon Bonds (ZCB)



- The term structure of interest rates was relatively flat in September 2018, as compared to September 2004 and 2008.

Additional Points: Zero-Coupon Bonds (ZCB)

- The price of a pure discount bond with maturity T per dollar of face value is:

$$B_0(T) = \frac{1}{(1 + r_0(T))^T}$$

where:

- $r_0(T)$ is the time-zero, rate of return on a T-period investment.
- **Question:** Why Do We Care About Zero-Coupon Bonds?

Additional Points: Zero-Coupon Bonds (ZCB)

- If we know the prices of pure discount bonds with maturities 1, 2, 3,... then we also know what the price of a risk-free bond with **any** complicated stream of future payments has to be.
- **Example:**
 - Suppose that $B_0(1) = \$0.91$, $B_0(2) = \$0.79$ and $B_0(3) = \$0.65$.
 - What is the price of a new risk-free bond with a face value of \$1,000 that makes annual coupon payments each period at a rate of 8% and matures in 3 periods?
 - The new bond makes payments of \$80 at $t = 1$, \$80 at $t = 2$ and \$1080 at $t = 3$.
- There are **two** possible approaches for solving this problem...

Additional Points: Zero-Coupon Bonds (ZCB)

Approach I

- The same stream of payments is made by a portfolio that includes:
 - 80 Pure Discount Bonds (PDBs) at $t = 1$.
 - 80 PDBs at $t = 2$.
 - 1080 PDBs at $t = 3$.
- Therefore, the price of this new bond must be the same as the price of the portfolio:

$$\text{Portfolio Value} = 80 \cdot (0.91) + 80 \cdot (0.79) + 1080 \cdot (0.65) = \text{\$838.00}$$

- **Bottom Line:** This approach is nothing more than using the discount bond prices directly.

Additional Points: Zero-Coupon Bonds (ZCB)

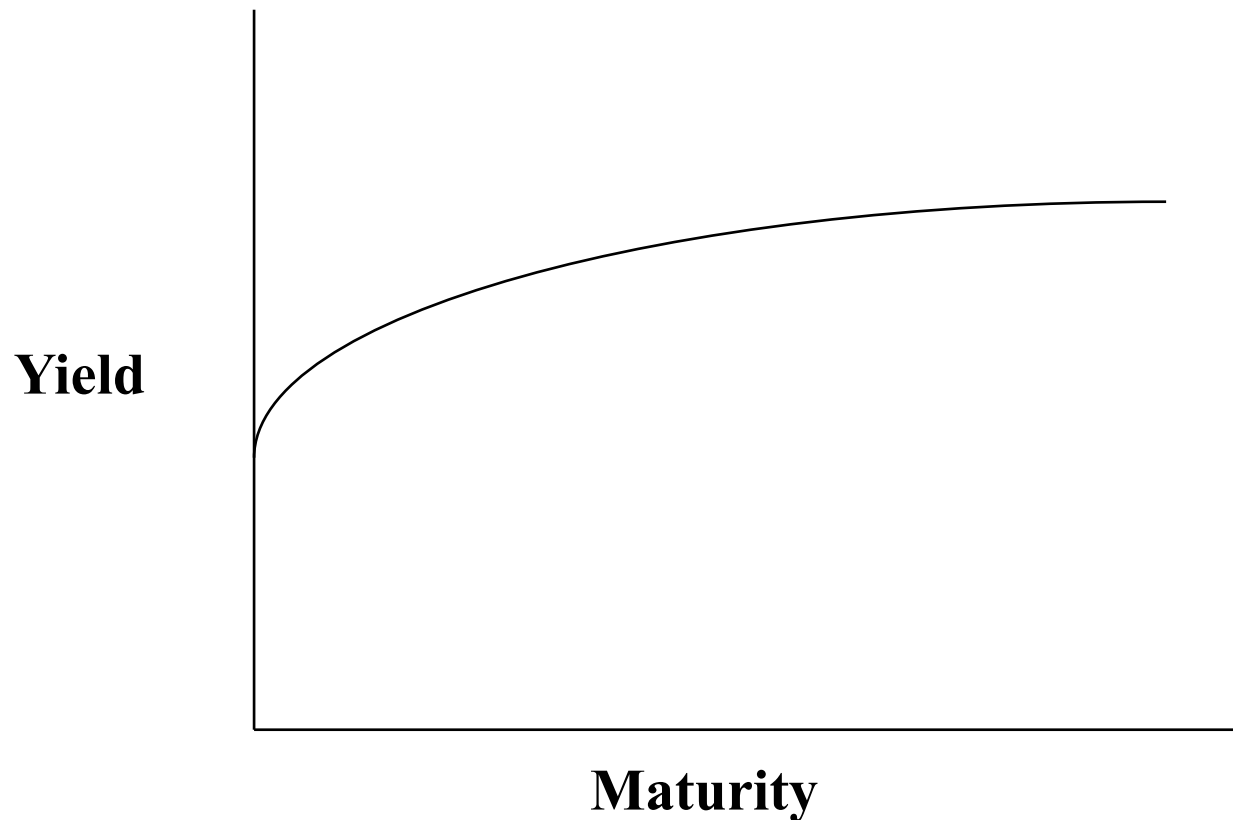
Approach II

- An equivalent approach is to solve for each interest rate first using the definition of $B_0(t)$. We obtain the following results:
 - $r_0(1) = 10\% \sim 0.098$
 - $r_0(2) = 13\% \sim 0.125$
 - $r_0(3) = 15\% \sim 0.154$
- Then we use these interest rates to compute the PV of the stream of payments associated with the bond:

$$\text{Bond Price} = \$80/1.10 + \$80/(1.13)^2 + \$1080/(1.15)^3 = \text{\textcolor{red}{\$838.00}}$$

Additional Points: Zero-Coupon Bonds (ZCB)

- The interest rates we just calculated can be graphed to obtain a spot yield curve, generally of the following form:



Additional Points: Zero-Coupon Bonds (ZCB)

- **Example:** Say that the three year spot yield curve is: $r_0(1) = 11\%$, $r_0(2) = 12\%$, $r_0(3) = 12.14\%$, and that you are considering a 3-year 10% annual coupon bond that has annual payments.
- The price of this bond is \$95 per \$100 face value:

$$95 = 10/1.11 + 10/(1.12)^2 + 110/(1.1214)^3$$

- The yield to maturity of this bond is 12.08%, which solves

$$95 = \frac{10}{(1 + YTM)} + \frac{10}{(1 + YTM)^2} + \frac{110}{(1 + YTM)^3}$$

- The yield to maturity tells you the average return over the life of this investment, but is **not** useful for pricing other similar bonds (e.g., a 3-year bond with a 7% coupon rate).

Additional Points: Zero-Coupon Bonds (ZCB)

- Another interpretation of the same result is that the value of a Coupon Bond, $B(C,T)$, must be equal to the value of a portfolio of zero-coupon bonds that gives the same cash-flow:

$$B(C, T) = \sum_{t=1}^T C \times B(0, t) + F \times B(0, T)$$

- **Note:** This is an **arbitrage-free relationship**.
 - If this relation were not satisfied, it would be possible to replicate the coupon bond with a portfolio of zeros and generate risk-free profits.
 - There would be an **arbitrage opportunity**!

Appendix: Implied Forward Rates

Implied Forward Rates

- Explaining the differences in returns for risk-free investments having different maturities is the same as explaining the **slope** of the yield curve at each point.
- Suppose you want to lend \$1 for 2 periods. You can either:
 - Buy a bond that matures in two periods.
 - Purchase two bonds in succession each with a one period horizon.
- The per-period return that you earn:
 - On the two-period bond is $r_0(2)$
 - On the first of the one-period bonds is $r_0(1)$.
 - We do not know what the return on the second one-period bond $r_1(1)$ will be until time 1.
- **Bottom Line:** $r_1(1)$ is called the **implied forward rate!**

Implied Forward Rates

- The interest rate on the second one-period investment that equates the payoffs of both strategies is the implied forward rate.

$$\$1 (1 + r_0(2))^2 = \$1 (1 + r_0(1)) (1 + f_1(1))$$

- There are lots of implied rates.
 - However, we only care about those that relate to **one-period** horizon investments.
 - From the time-0 spot yield curve, we compute the implied forward rates from:

$$(1 + r_0(T))^T = (1 + r_0(T-1))^{T-1} (1 + f_{T-1}(1))$$

- **Question:** Why do we care about the implied forward rate?

Shape Of The Yield Curve

- Implied forward rates help explain the **shape** of the yield curve.
 - They provide an **analytical** connection between adjacent points on the yield curve.
- If $f_{T-1}(1) > r_0(T-1)$, then the yield curve slopes **up** between horizons $T-1$ and T ;
- If $f_{T-1}(1) < r_0(T-1)$, then the yield curve slopes **down** between horizons $T-1$ and T .
- **Bottom Line:** Explaining differences in returns on investments having different horizons is equivalent to explaining differences in implied forward rates.

Measuring The Term Structure

- **Problem:** The great majority of bonds pay coupons, and most available data pertain to coupon bonds, so we must develop a general approach to calculate spot and forward rates from coupon bonds.
- **Trick:** Treat each coupon payment as a separate “mini”-zero coupon bond.
 - A coupon bond then becomes just a portfolio of many zeros with different maturity dates.
- **Example:** Suppose that we observe an 8% coupon bond making annual payments with two years until maturity selling at \$1,048.36, and a 10% coupon bond, also making annual payments with two years to maturity, selling at \$1,085.41. Both bonds have par value of \$1,000. What are the spot rates?

Measuring The Term Structure

- Let $B_0(1)$ and $B_0(2)$ be the discount factors or the ZCB prices. By no arbitrage:

$$\$1,048.36 = B_0(1) * \$80 + B_0(2) * \$1,080$$

$$\$1,085.41 = B_0(1) * \$100 + B_0(2) * \$1,100$$

$$B_0(1) = \mathbf{0.9524} \qquad B_0(2) = \mathbf{0.90016}$$

- The implied zero coupon curve is given by:

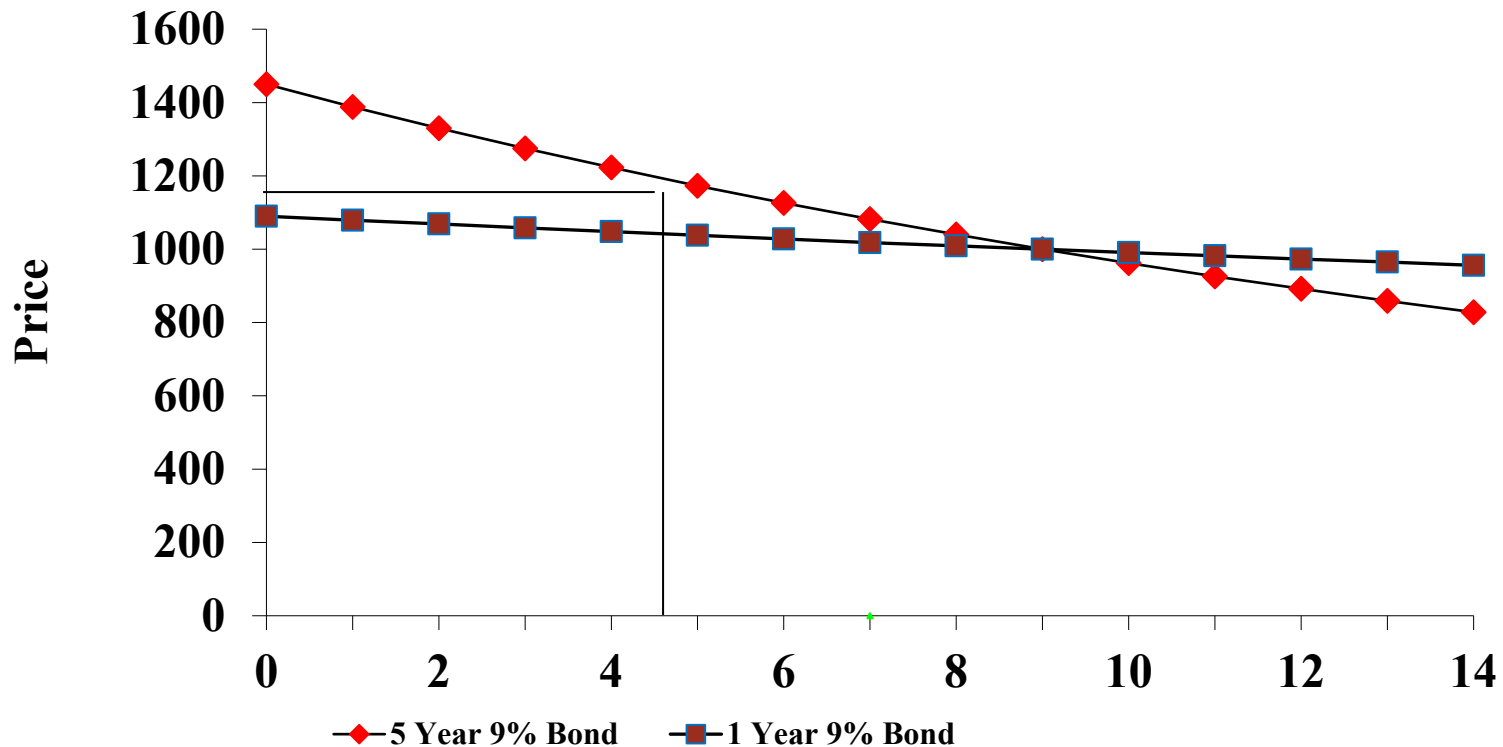
$$r_0(1) = 1/0.9524 - 1 = \mathbf{5\%}$$

$$r_0(2) = (1/0.90016)^{1/2} - 1 = \mathbf{5.4\%}$$

Appendix: Volatility, Duration, and Convexity

Volatility and Duration

- Any instrument with longer duration implies that cash-flows are more **back-loaded**.
 - Longer duration instruments are **more** interest rate sensitive.
 - Long-term bonds tend to be **more** sensitive to interest rate risk.



Volatility and Duration

- Duration is useful mainly since the percentage change in the price of a bond due to a change in the bond's yield to maturity is proportional to the duration times the change in the yield:

$$\% \text{ Change in Bond Prices} \approx -D \times \frac{\% \text{ Change in Yield (\% Points)}}{1 + Y}$$

$$\% \text{ Change in Bond Prices} \approx -D^M \times \% \text{ Change in Yield (\% Points)}$$

where:

- y is the level of the yield curve before any changes.
- D^M is called the **modified duration**. $D^M = D/(1 + y)$
- The minus sign in front of D reflects the fact that when the yield curve rises, the bond price falls.
- Remark:** The longer a bond's duration, the **greater** is its volatility.

Volatility And Duration: Examples

- **Example I:** Consider a 2-year 8% coupon bond with face value \$100 with a current YTM = 5%. By how much does the price of this bond change if the yield curve moves up by 1%?

- The bond value is the sum of the PV's of the two cash flows:

$$\text{Bond Price} = \frac{\$8}{1.05} + \frac{\$108}{(1.05)^2} = \$7.62 + \$97.96 = \$105.58$$

- To get the **duration** of this portfolio, we weight the years by the present value of the cash flows divided by the bond value:

$$D = (1 \text{ year}) \times \frac{\$7.62}{\$105.58} + (2 \text{ years}) \times \frac{\$97.96}{\$105.58} = 1.9278 \text{ years}$$

Volatility And Duration: Examples

- The **modified duration** is therefore:

$$D^M = \frac{1.9278 \text{ years}}{1 + 0.05} = 1.836 \text{ years}$$

- Thus, if the yield curve moves up by 1 percentage point, this bond price drops approximately 1.836% in price.

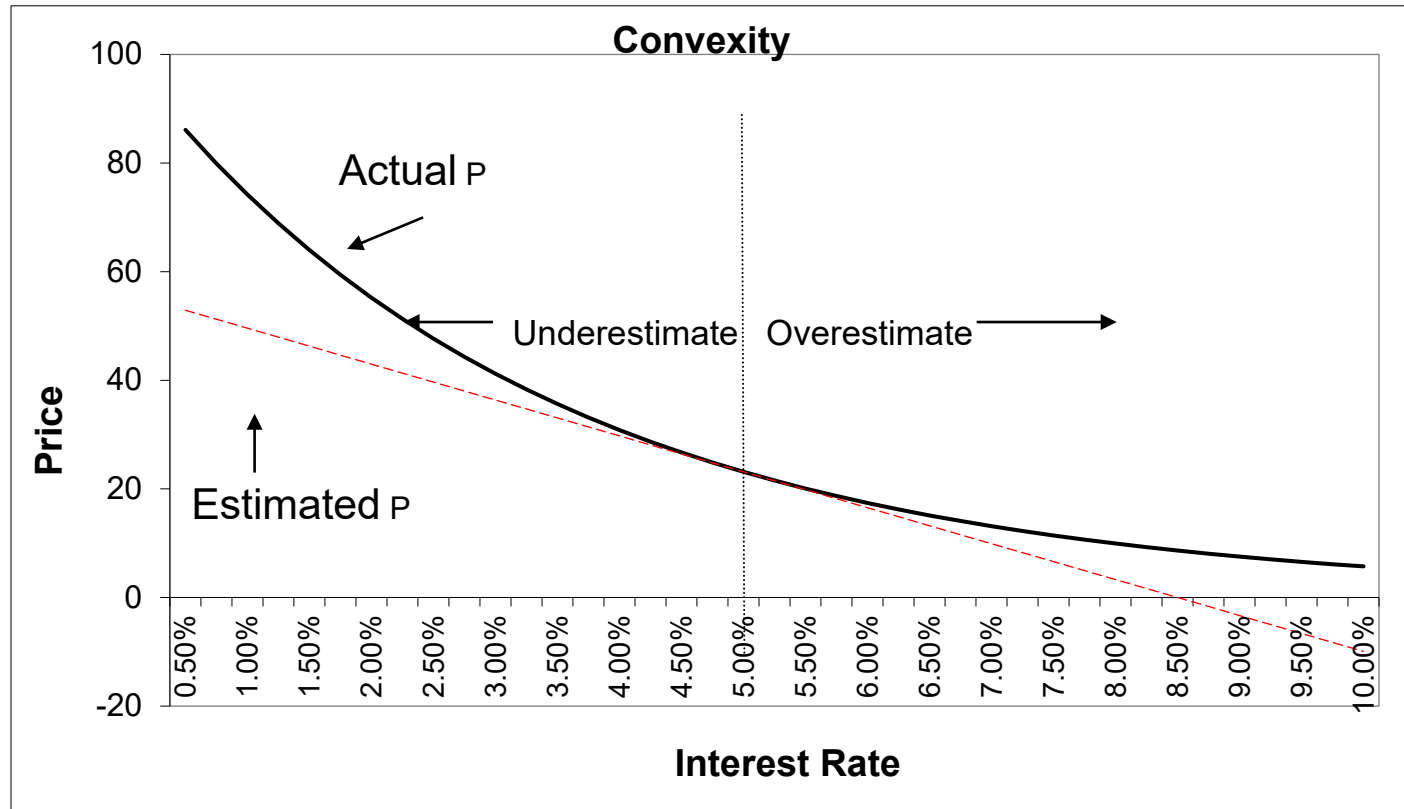
$$\% \text{ Change In Bond Prices} = -1.836\% \times 1 = -\mathbf{1.836\%}$$

- **Example II:** The yield-to-maturity on a 20-year 5% coupon bond with face value \$1,000 is currently 5%. Find this bond's duration and modified duration, and calculate how much the price of this bond would increase (in %) if its yield fell to 4.5%

Do this in Excel, either using the DURATION function or by setting up the calculation based on the definition of duration.

Convexity Property

- The interest rate **sensitivity** is given by the percentage change in price per percentage change in interest.



- The bond's volatility is simply the slope of the line relating the bond price to the interest rate.
 - The bond's volatility changes as the interest rate changes.

The Spread

